



# **Toegepaste Micro-economie: Public economics**

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Week 5:  
Taxation: incidence and efficiency

# 5. Taxation: incidence and efficiency

Right to tax very important for functioning governments

Relating to previous weeks, tax revenue needed to:

- Finance public provision of goods
- Finance anti-poverty measures → redistribution
- Finance (Pigouvian) subsidies
- Of course, also ‘input cost’ government

So far, no systematic analysis of the effects of taxation  
(apart from Pigouvian tax)

# 5. Taxation: incidence and efficiency

Evaluation / analysis of taxes (and tax systems):

As always: look at **efficiency** and **distribution**

Efficiency:

- Minimise distortionary effects (effects on consumption, investment, labour supply) → ‘neutrality’
- In case of externalities: correct market failure (Pigou)

Distribution (or incidence) of tax burden:

- Who suffers, and how much
  - Horizontal equity: ‘equal’ people pay equal taxes
  - Possibly: vertical differentiation
  - People who are ‘better off’ pay more taxes

# Today's roadmap

1. General remarks and some figures
2. **Distribution** of tax burden: 'incidence'
  - Statutory versus economic incidence
  - Tax shifting
  - Partial equilibrium
3. **Efficiency** effects
  - Tax distortions → 'Excess Burden'
  - Individual and market level

Next week: Optimal taxation

More in-depth: B3-course "Public Finance"

# 5. Taxation: incidence and efficiency

<https://ourworldindata.org/taxation>

Tax levels and main sources of tax revenue differ widely across countries and time

Sources:

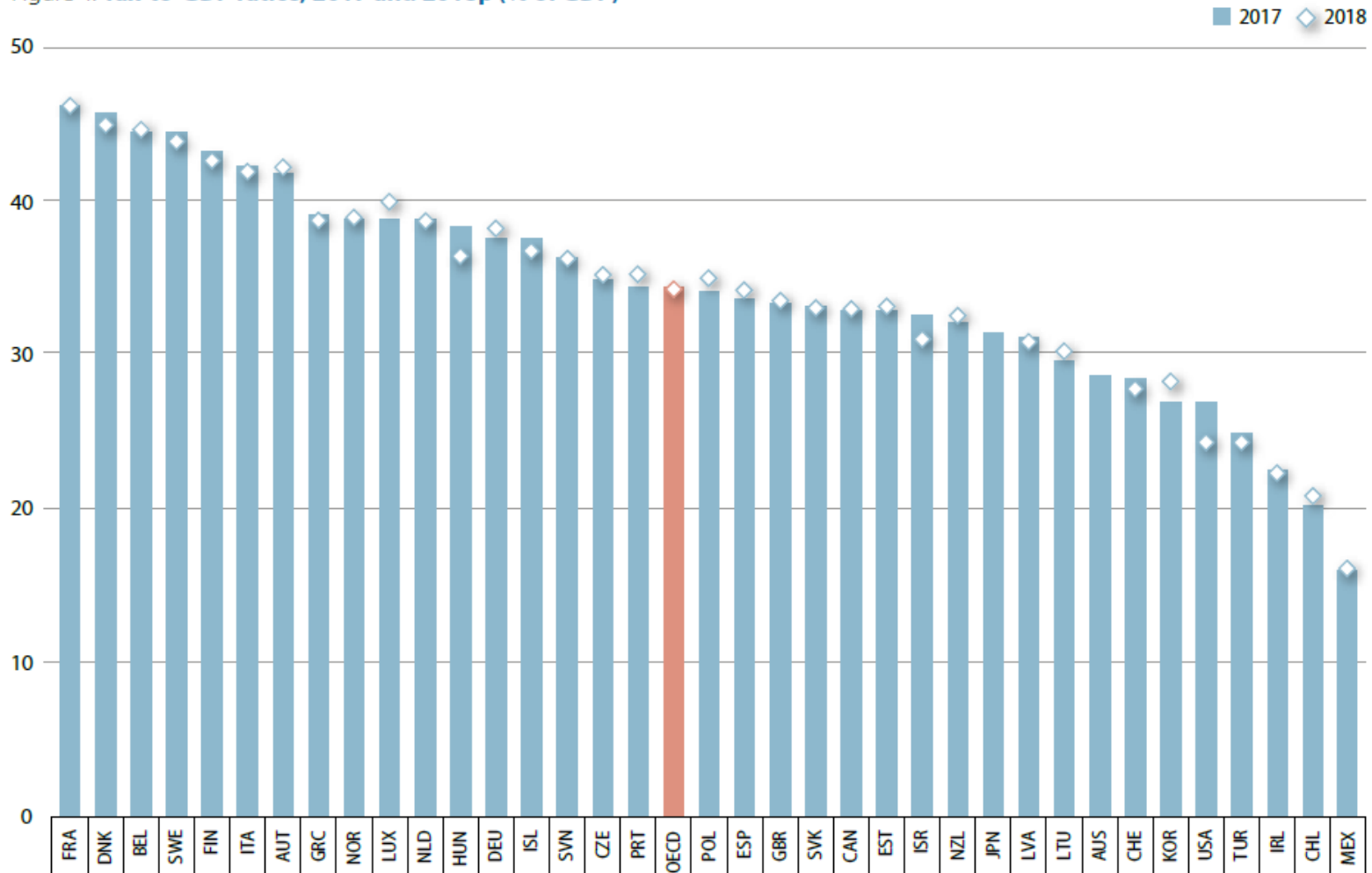
Mixture of taxes on income, consumption and wealth, country-specific and partly historically determined

Common trend over time in OECD countries:

- First excises (for instance on sugar)
- In 20<sup>th</sup> century: shift to income tax, very progressive
- Latest trends: more consumption tax (VAT), decrease in both average and marginal income tax rates.

# Tax level OECD countries

Figure 4. **Tax-to-GDP ratios, 2017 and 2018p (% of GDP)**

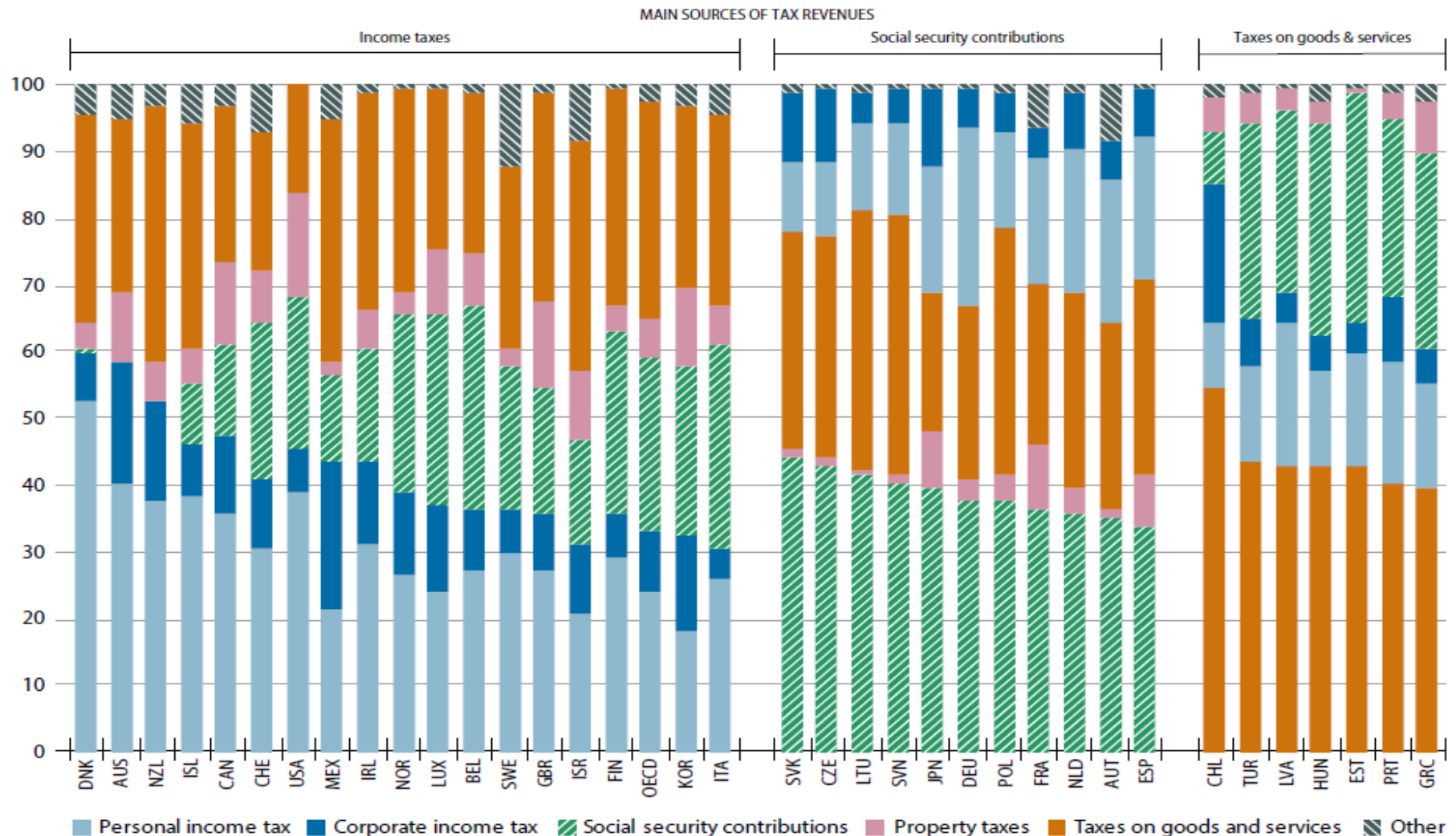


Notes: Preliminary data for 2018 were not available for Australia and Japan.

Source: Data from Revenue Statistics 2019, <http://oe.cd/revenue-statistics>

# Sources of tax revenue OECD countries

Figure 5. Tax structures, 2017 (% of total tax revenue)



Note: Countries are grouped and ranked by those where income tax revenues (personal and corporate) form the highest share of total tax revenues, followed by those where social security contributions, or taxes on goods and services, form the highest share.

Source: Data from Revenue Statistics 2019, <http://oe.cd/revenue-statistics>

# Taxation: general remarks

1. Only people can bear tax burden  
→ Tax on ‘firms’ or ‘capital’ paid by owners!
2. Unit tax versus ad valorem tax:  
Ad valorem is levied as percentage of price (or wage etc.)  
→ VAT is ad valorem; excise is unit tax

# Taxation: general remarks

## 3. Distinguish average and marginal tax rate

Example income tax: first 5000 euro no tax, 25% tax on income  $Y$  over 5000 euro.

$$\rightarrow T = 0.25(Y - 5000)$$

Suppose income is 25.000 euro. Then:

- Total tax paid:  $T = 0.25(25000 - 5000) = 5000$
- Average tax rate:  $T / Y = 5000 / 25000 = 0.2$
- Marginal tax rate: 0.25

# Taxation: general remarks

4. Progressiveness measured by **average** tax rate

→ If average tax rate:

- Increases in income, tax system is progressive
- Decreases in income, tax system is regressive
- Is constant in income, tax system is proportional

‘Universal basic income’ vs ‘negative income tax’  
(See exercise 14.8)

<https://ourworldindata.org/taxation>

# Incidence

Who pays taxes versus who bears tax burden

Who pays: statutory incidence

- VAT: Consumers
- Dutch wage tax: partly employers, partly employees

Who bears tax burden: economic incidence

- Which groups are made worse off by the tax?
- Not necessarily the party that pays!

**→ Tax shifting**

Example: auctioning emission permits

# Incidence

Economic incidence:

Imposing a tax affects all parties involved in production and consumption

→ Market forces determine who bears tax burden

Striking result:

**Economic incidence totally independent of stat. incidence**

# Tax in partial equilibrium

General remarks:

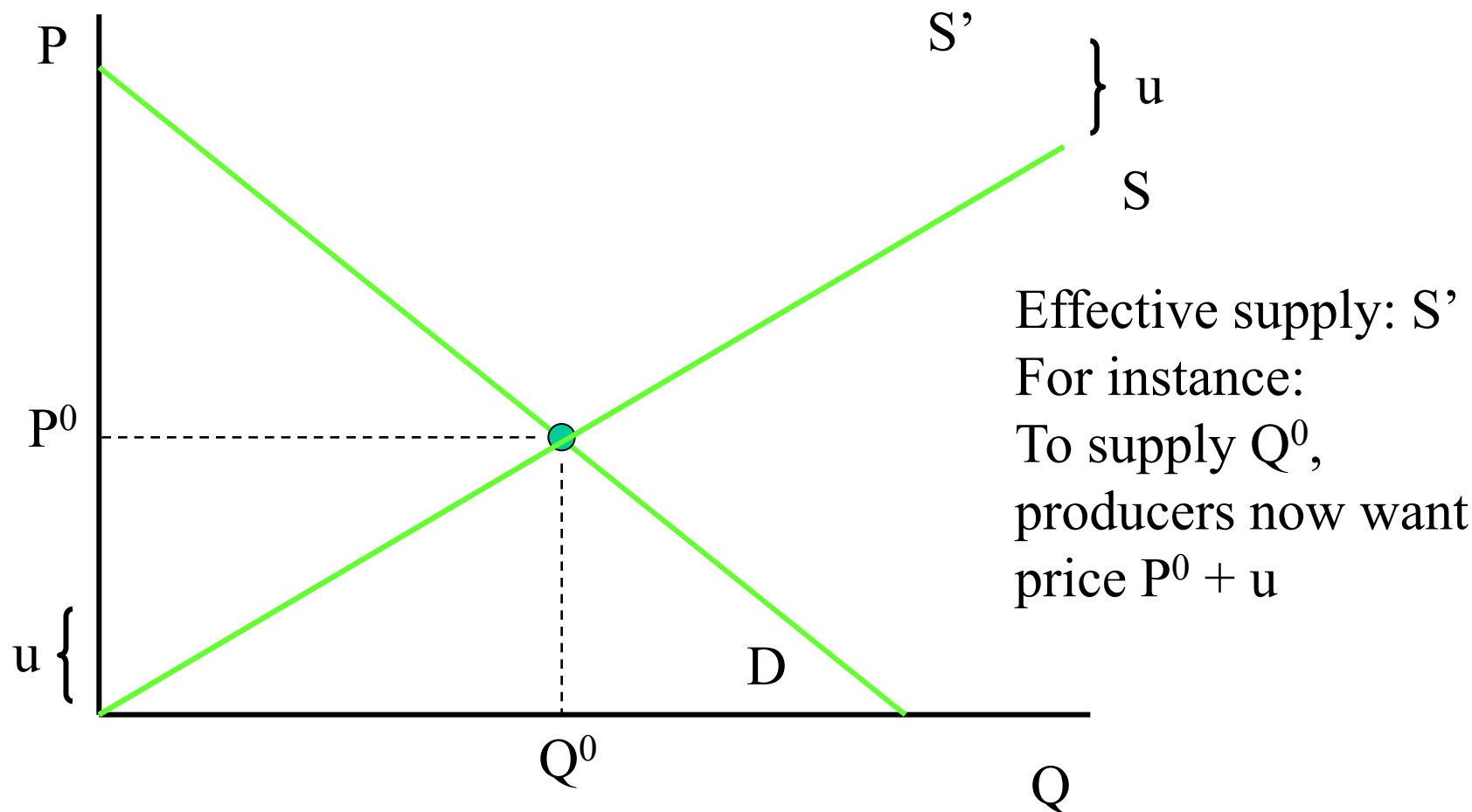
- Attention restricted to market where tax is imposed
- OK when tax is ‘small’ compared to economy
- Goal: to understand how a tax affects market equilibrium, total welfare, and distribution of tax burden

Tax in General Equilibrium:

- Covered in Rosen and Gayer, ch. 14
- Interesting, but not part of exam material

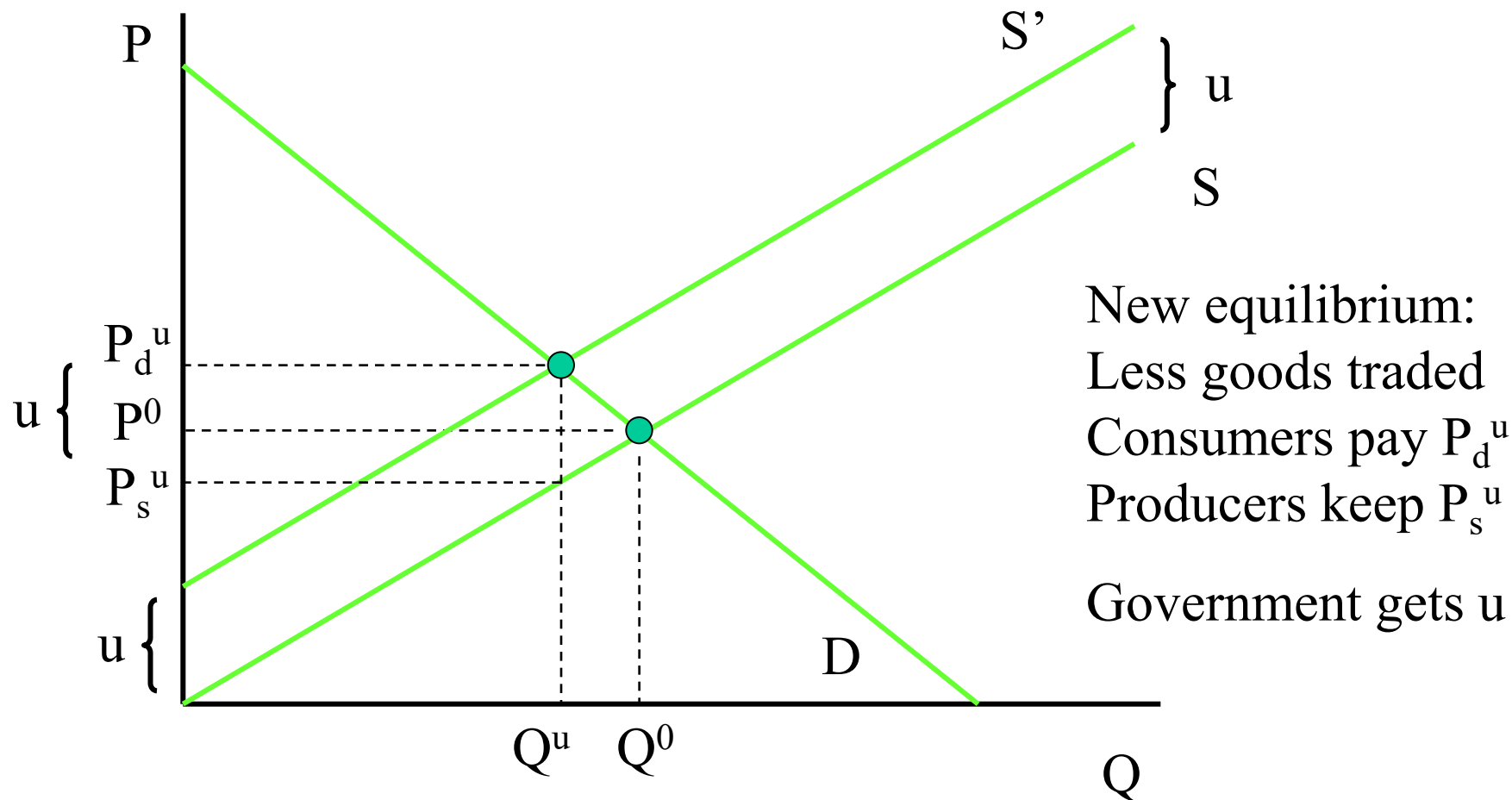
# Tax in partial equilibrium

Suppose unit tax  $u$  levied on producers



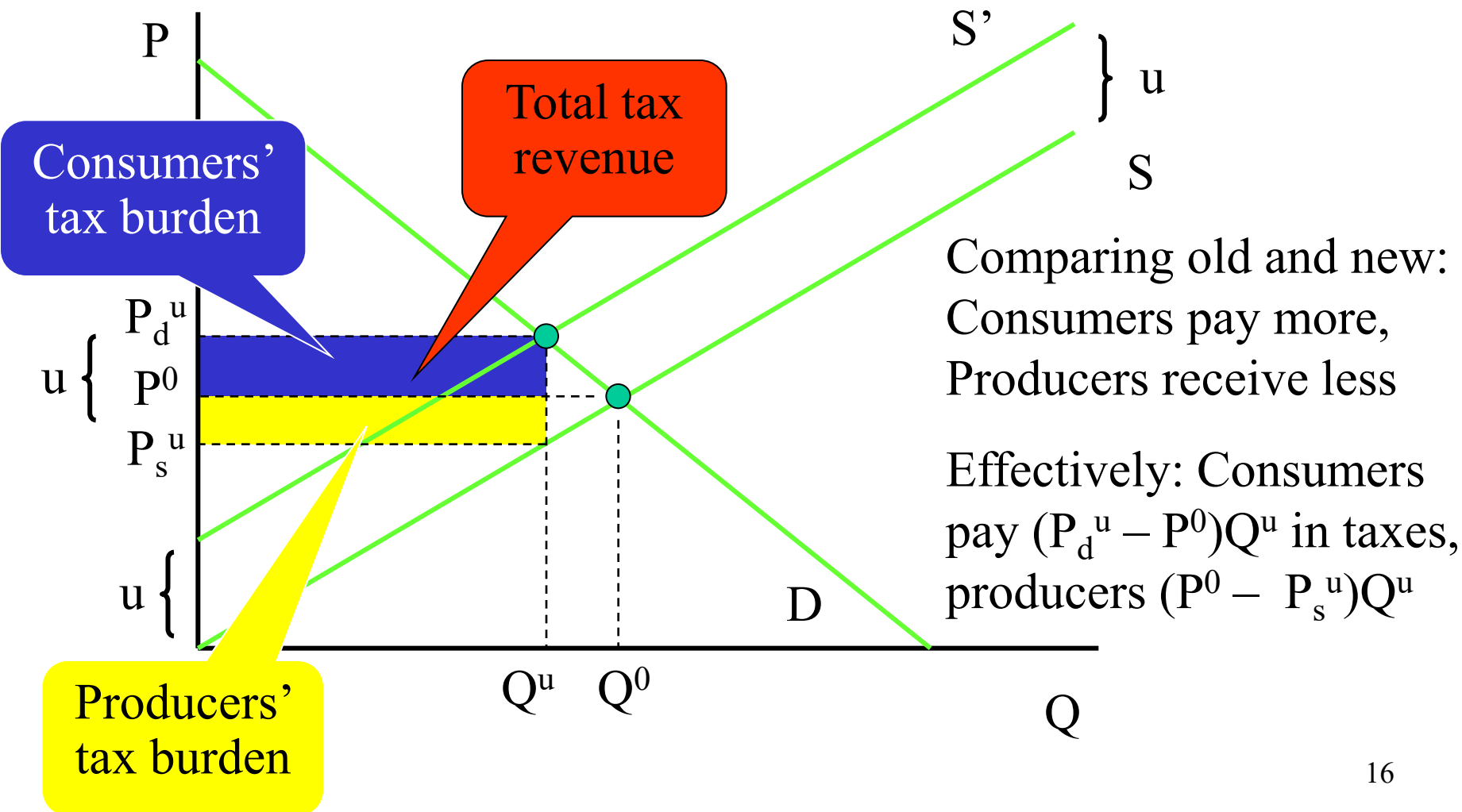
# Tax in partial equilibrium

New equilibrium with unit tax  $u$  levied on producers



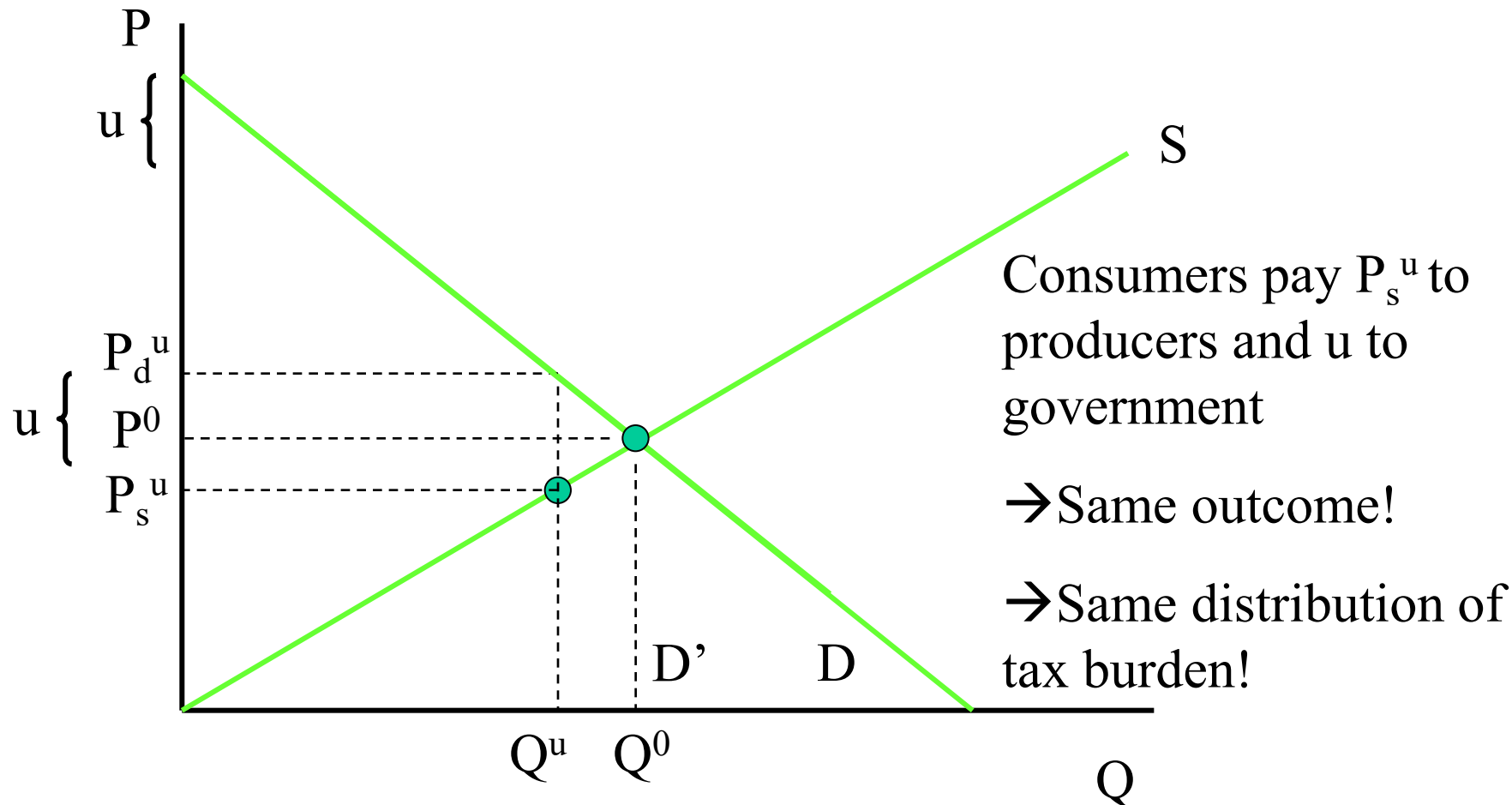
# Tax in partial equilibrium

## Distribution of tax burden



# Tax in partial equilibrium

Unit tax  $u$  levied on consumers yields equivalent outcome



# Tax in partial equilibrium

Mathematical ‘proof’ that outcome independent of who pays tax (for linear demand and supply) :  
→ Exercise 14.6

Intuitive explanation:

- Difference between consumer and producer price is unit tax:  $P_d = P_s + u$
- $Q_d$  decreases in consumer price
- $Q_s$  increase in producer price
- Only one point where  $Q_d = Q_s$  and  $P_d = P_s + u$

# Tax in partial equilibrium

What does determine the distribution of tax burden?

Crucial: (relative) **elasticities of demand and supply**

Recall elasticity (Frank ch. 4):

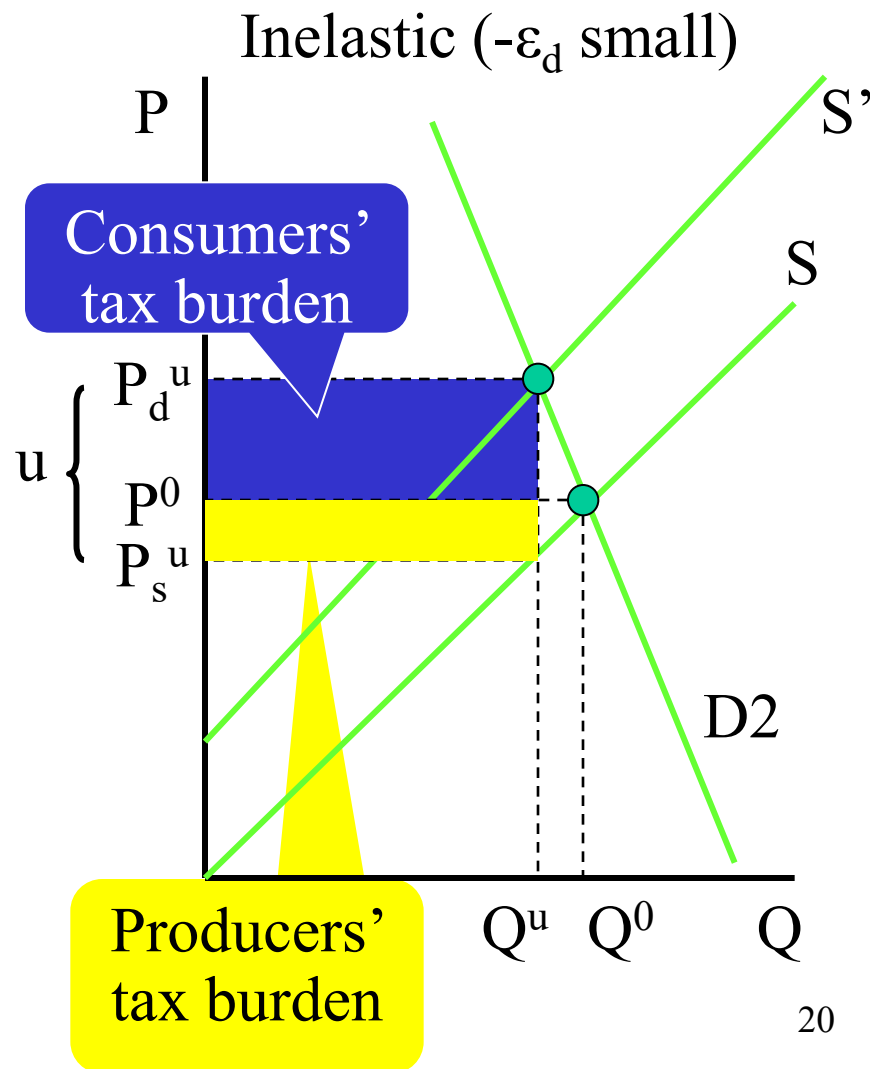
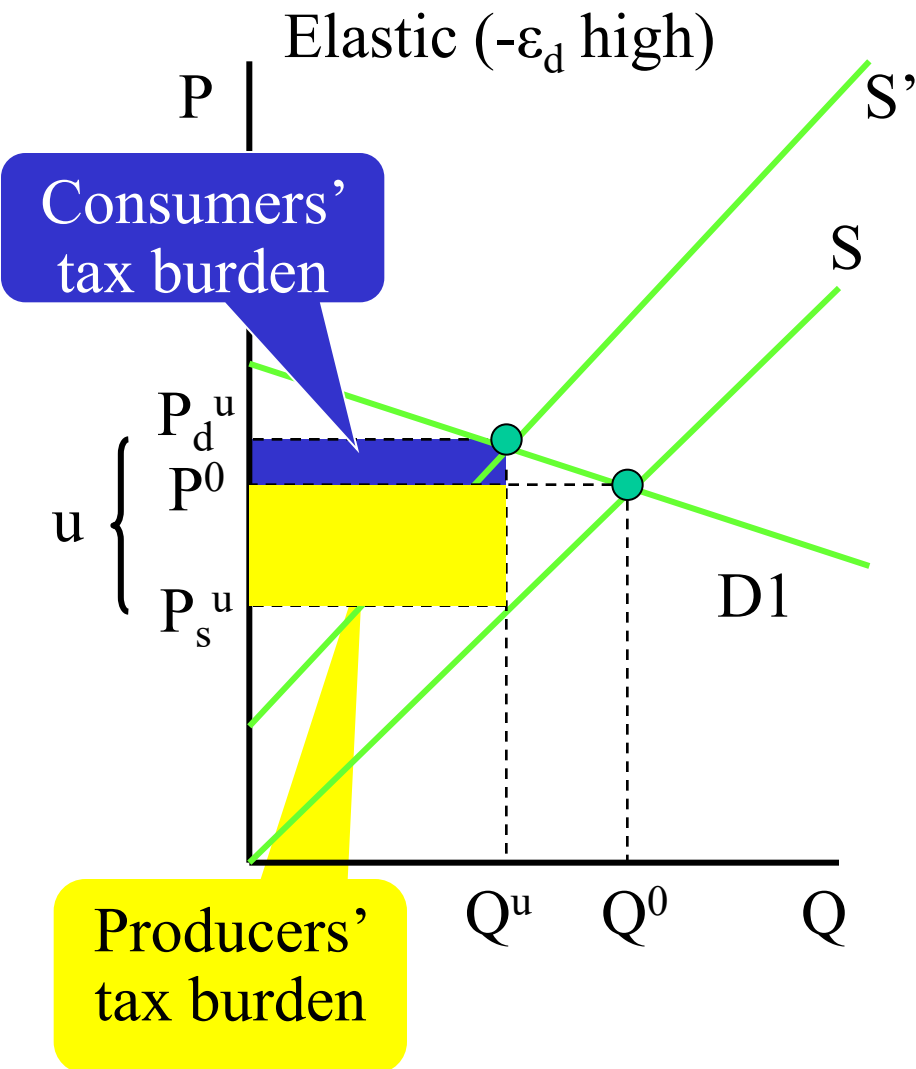
Percentage change in quantity demanded / supplied if price increases by 1 %

$$\varepsilon = \left| \frac{dQ}{dP} \frac{P}{Q} \right|$$

- Given demand elasticity, tax burden suppliers decreases in supply elasticity, and vice versa.
- In other words: the more inelastic, the higher the share of the tax burden

# Tax in partial equilibrium

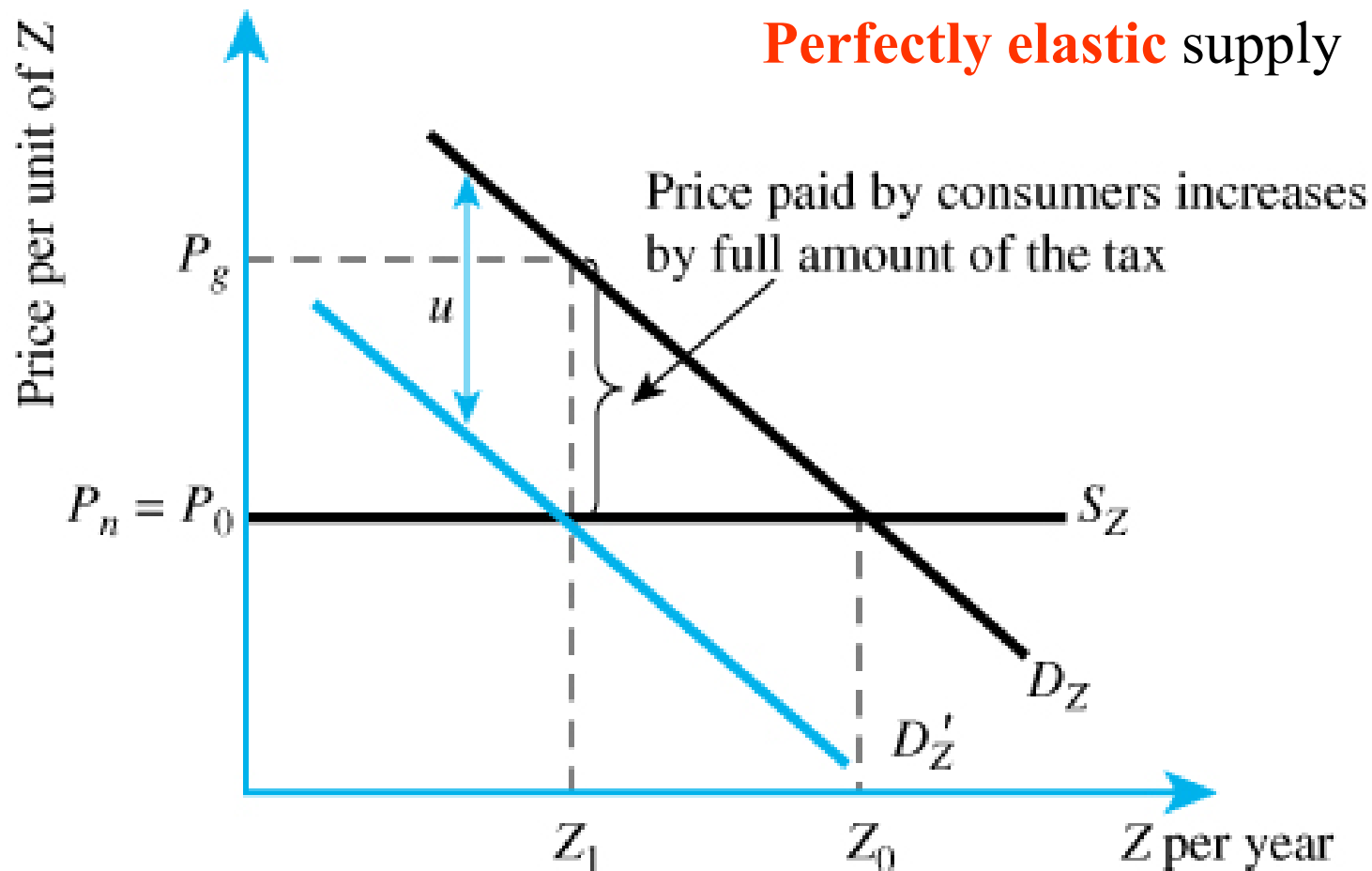
Distribution of tax burden: elastic versus inelastic demand



# Tax in partial equilibrium

Distribution of tax burden; the extremes:

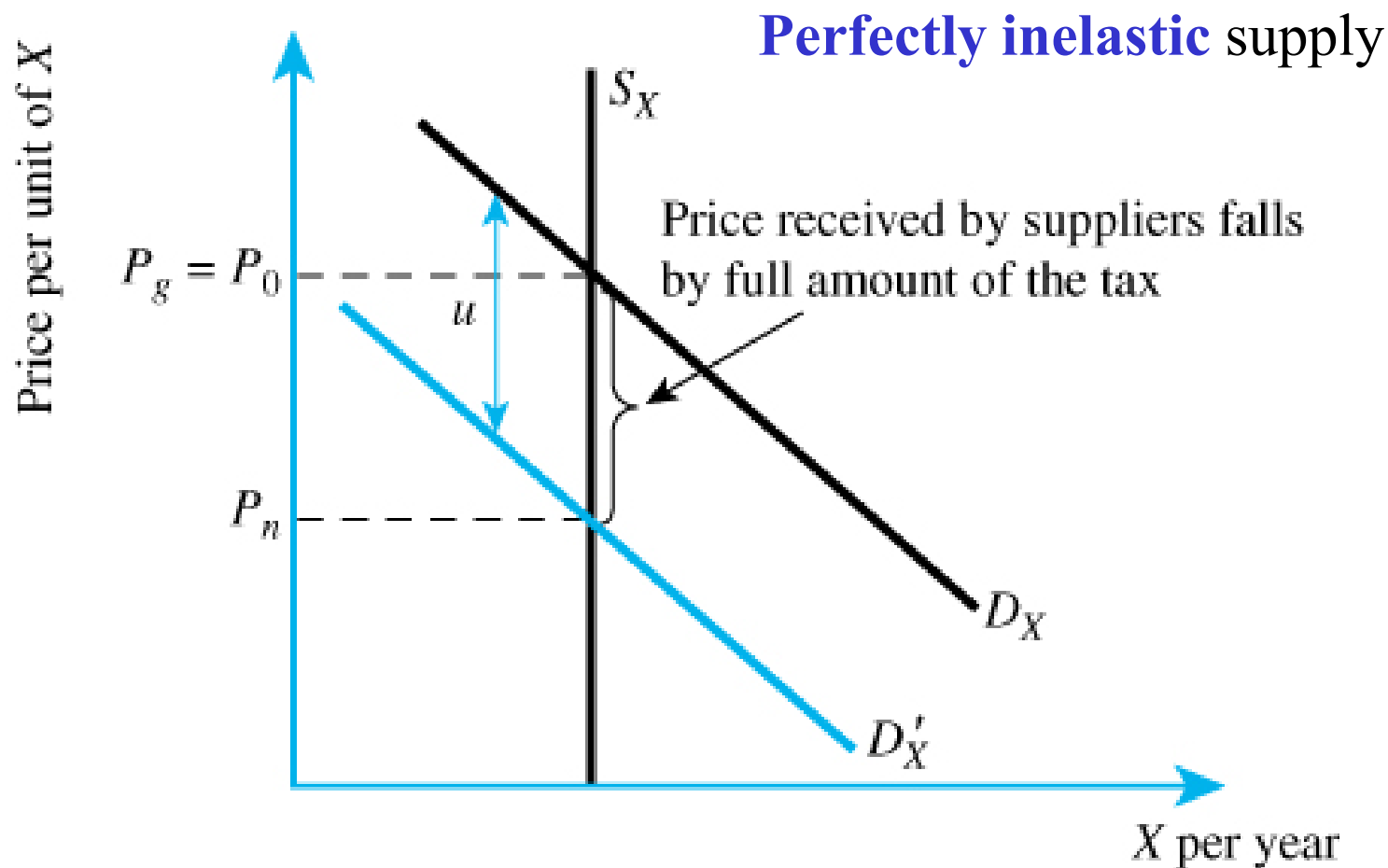
**perfectly elastic** supply and **perfectly inelastic** supply



# Tax in partial equilibrium

Distribution of tax burden; the extremes:

**perfectly elastic** supply and **perfectly inelastic** supply



# Tax in partial equilibrium

## Concluding remarks:

- Whether tax is levied on producers or consumers is economically irrelevant (in the long run)  
Frictions & switching cost may imply that in the short run, it does matter
- Same holds in input market: whether employers or employees have to pay payroll tax is irrelevant.
- Distribution of tax burden depends on (relative) elasticities of supply and demand.

# Incidence: Concluding remarks

Who pays taxes is irrelevant.

What matters is economic division of tax burden

Through changes in prices, everyone partially bears tax burden

Exact division crucially depends on elasticities....

→ The better the (non-taxed) alternatives (= the more elastic demand / supply), the smaller the share of the tax burden.

.... which can change over time

→ elasticities tend to be larger in long-run than in short-run

# Efficiency effects of taxation

Taxes paid by consumers and producers are not lost:

→ Tax revenue for the government

→ Transfers, do not in itself reduce efficiency

Generally, however, total loss in consumer and producer surplus is larger than tax revenue

Difference is called:

**Excess Burden**

(also called Deadweight Loss)

# Efficiency effects of taxation

Now:

Graphical analyses of excess burden of a tax

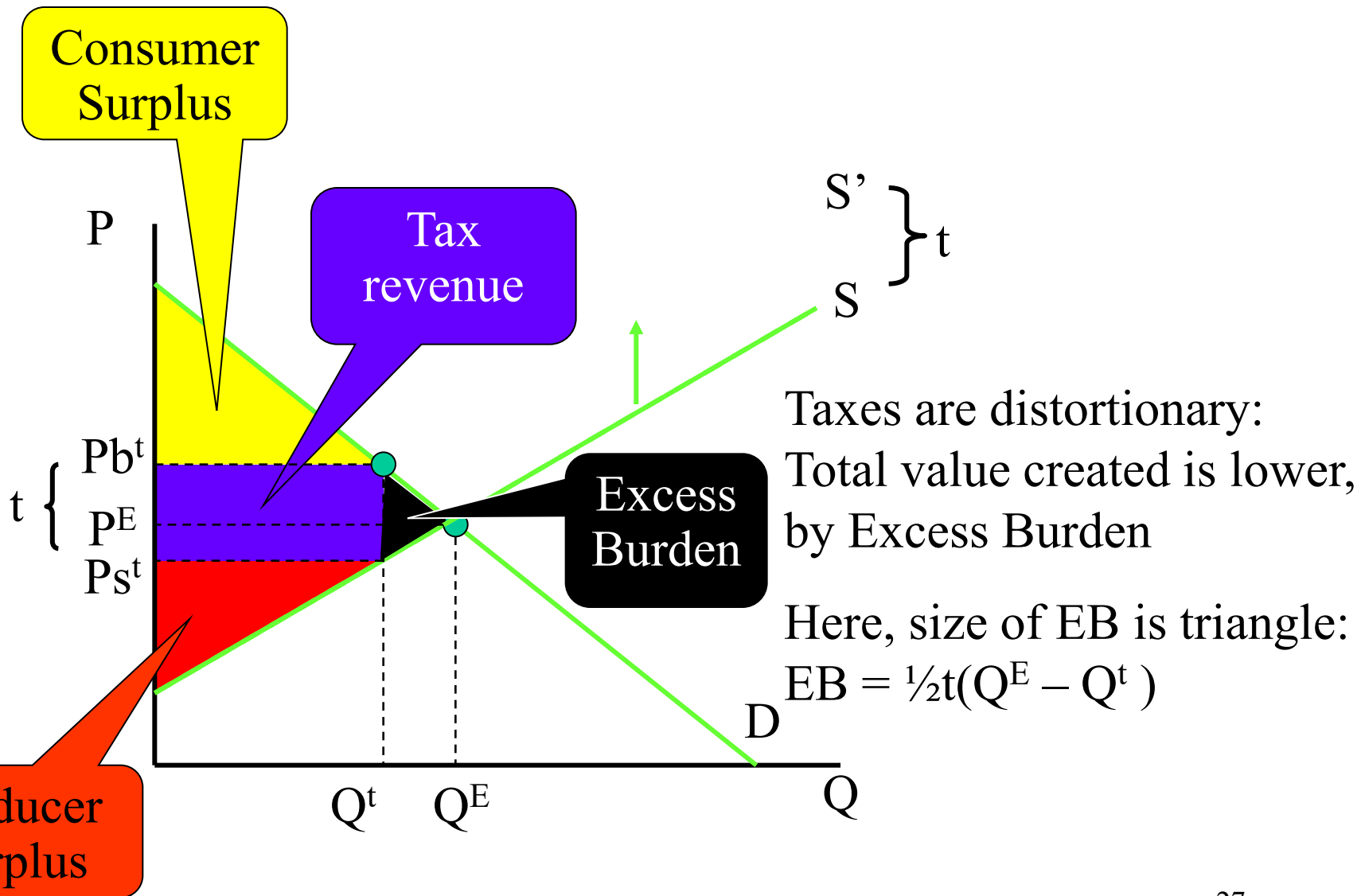
- Market level
- Individual level  
(note: in essence similar to part on work incentives last week!)

Next week:

Minimising excess burden: → Optimal taxation

# Efficiency effects of taxation: market

Excess burden of a unit tax  $t$



# Efficiency effects of taxation

Why does the Excess Burden arise?

Recall: Perfect competition with goods X and Y yields:

$$MRS = MRT = P^X / P^Y$$

Suppose suppliers bear share  $a$  of tax burden of unit tax  $t$  on X.

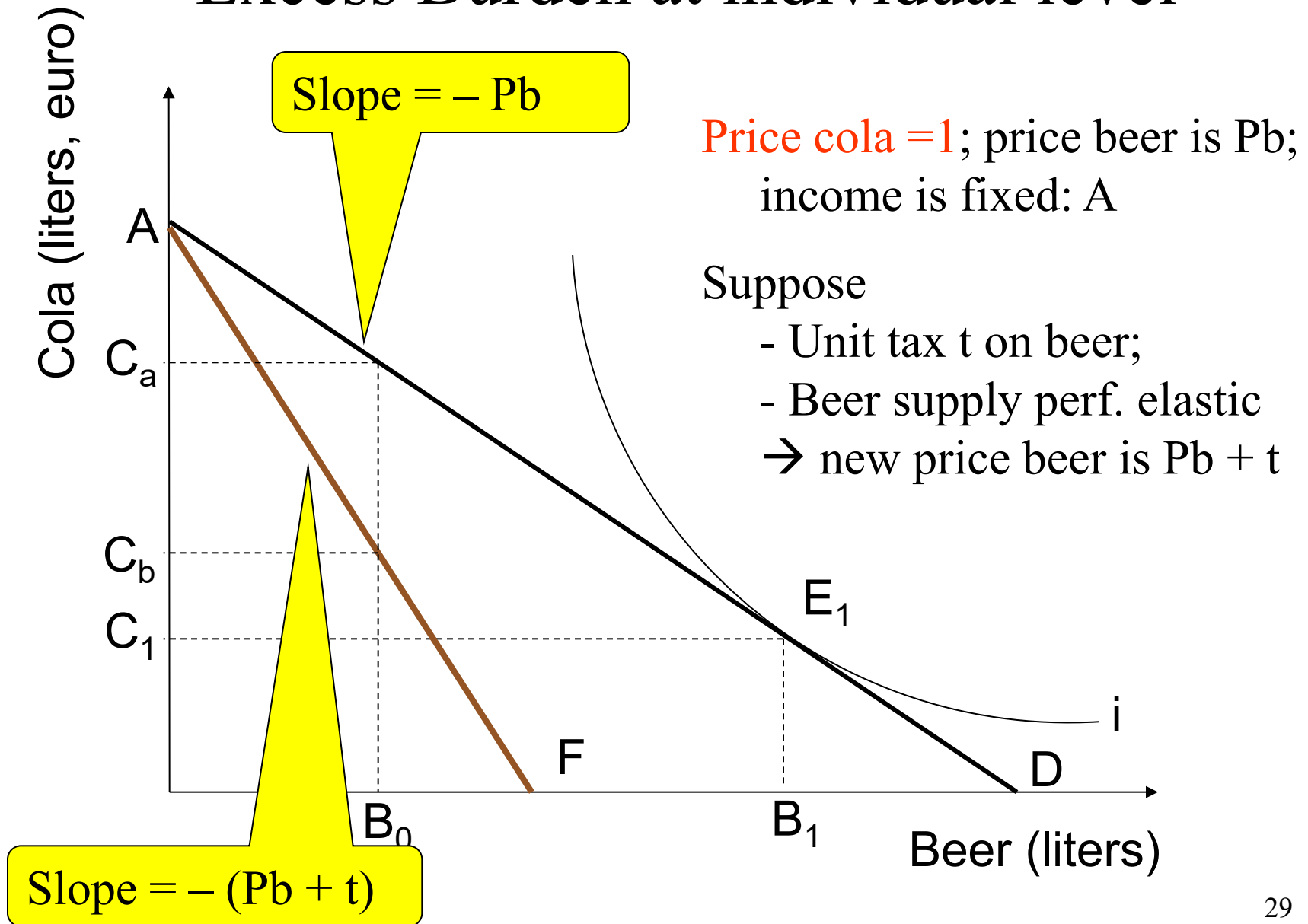
→ Suppliers keep  $P^X - at$ , consumers pay  $P^X + (1-a)t$

So: Consumers choose bundle such that  $MRS = \frac{P^X + (1-a)t}{P^Y}$

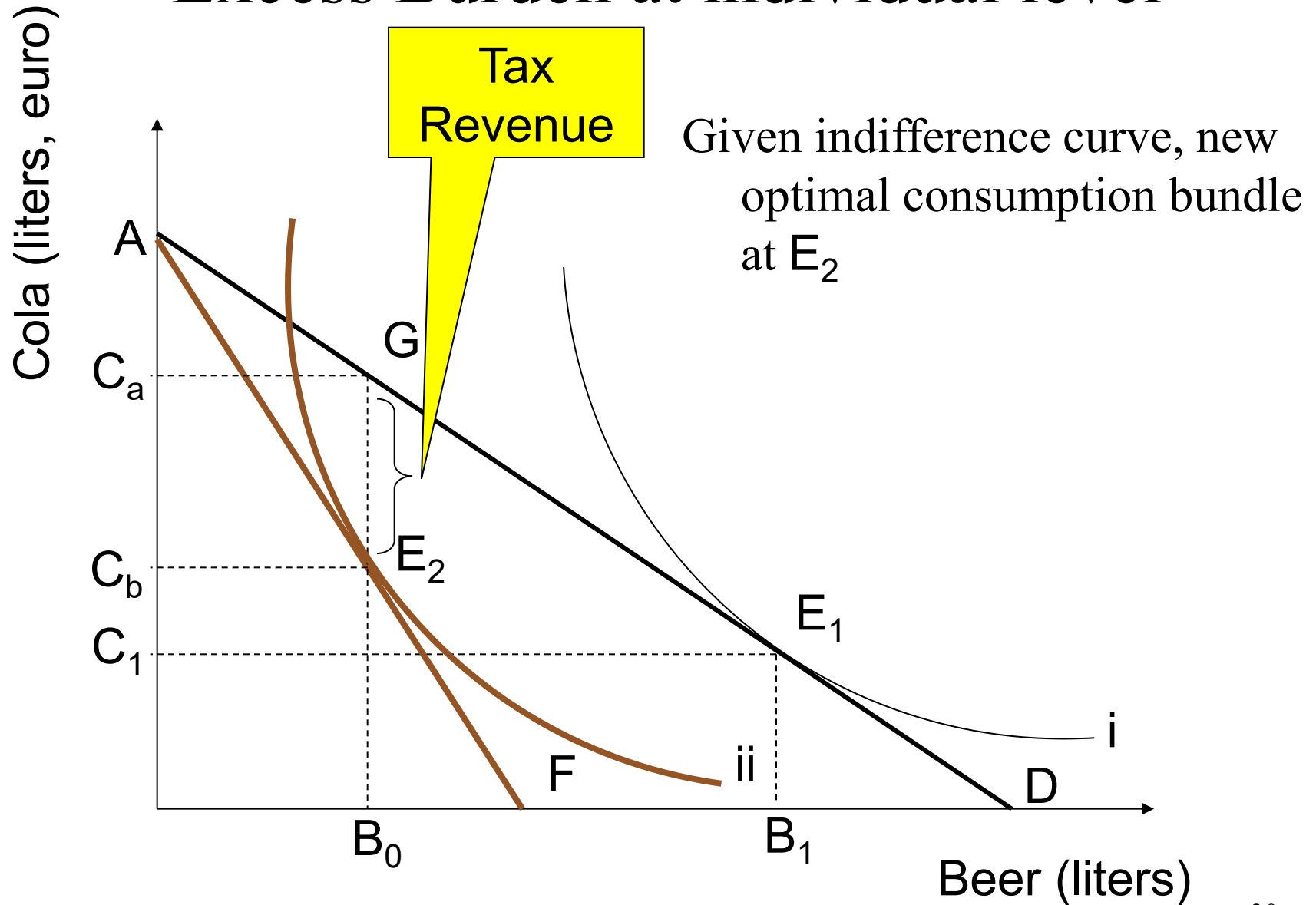
Producers use resources such that  $MRT = \frac{P^X - at}{P^Y}$

→  $MRS \neq MRT$  !

# Excess Burden at individual level



# Excess Burden at individual level



# Efficiency effects of taxation

Beer tax leads to utility loss

Question:

Before the tax, how much money could we take away to make consumer equally well off as after the beer tax?

→ Answer = **‘Equivalent Variation’**

If equivalent variation  $>$  tax revenue, then the utility loss  $>$  tax revenue

→ Tax leads to reduction in social welfare:

→ **Excess Burden**

# Efficiency effects of taxation

How to find EV:

Shift before-tax budget constraint down until it has a tangency-point with after-tax indifference curve

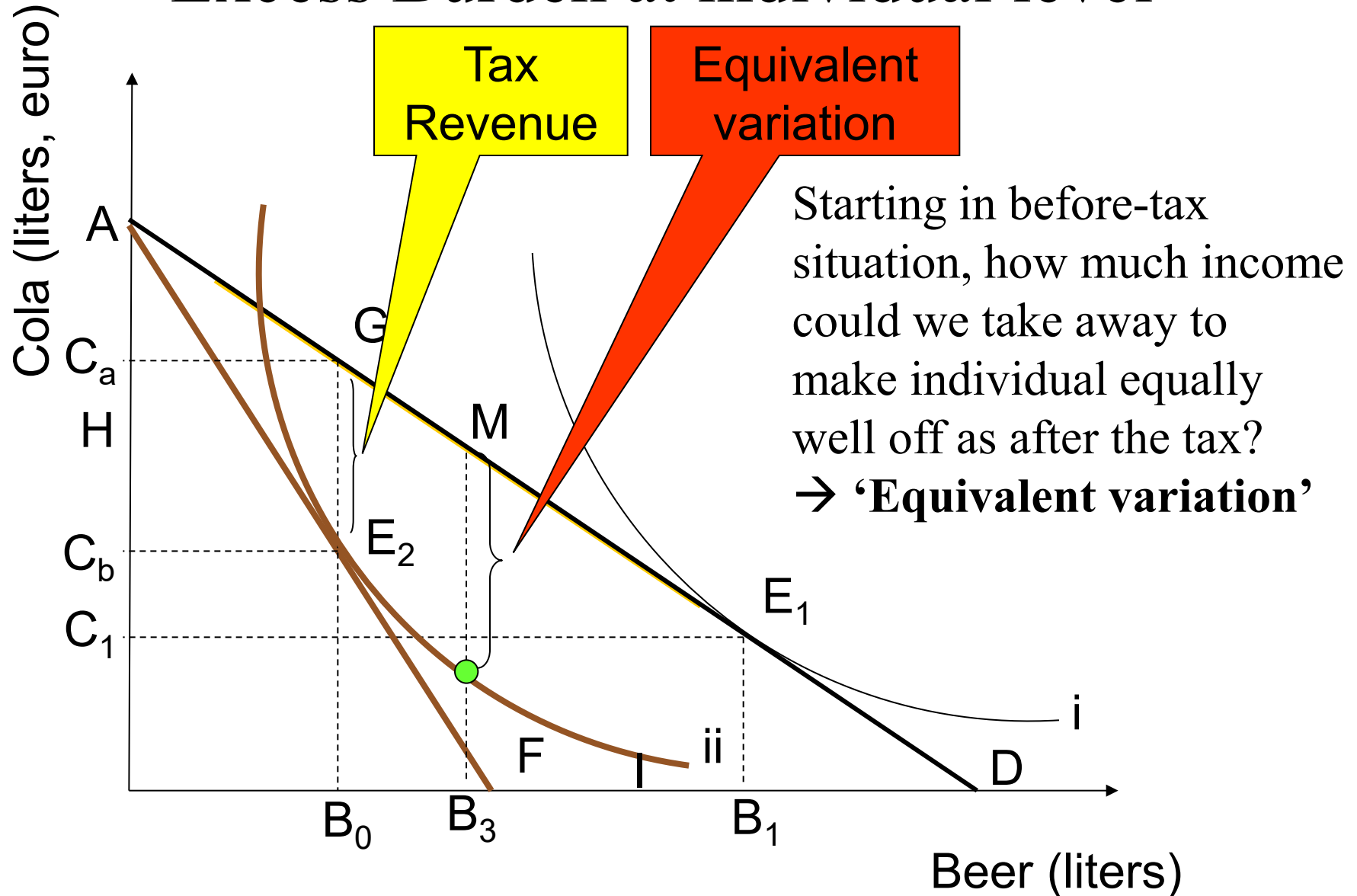
→ EV equals difference between old and shifted BC

Note:

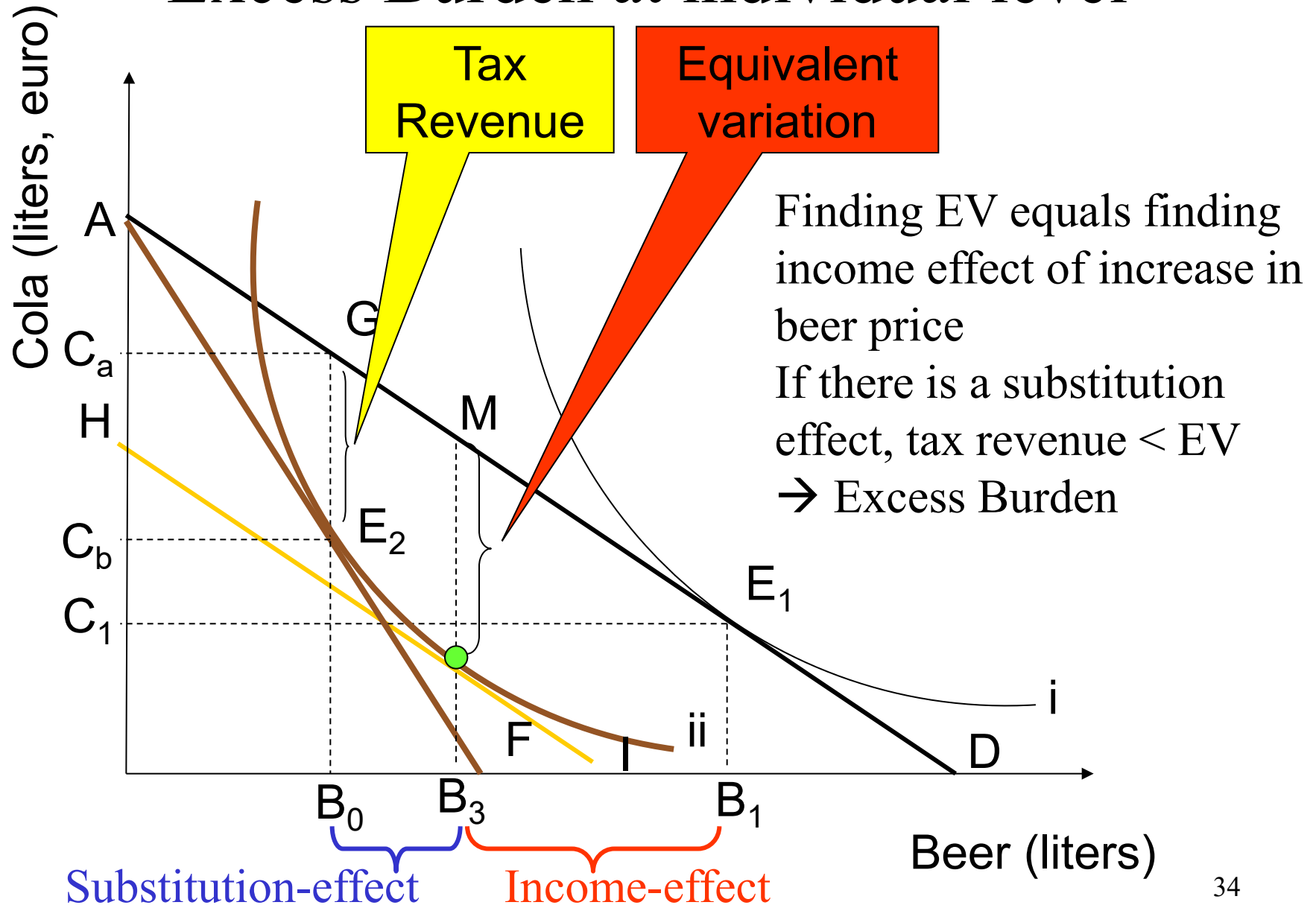
Effectively, we look for the income and substitution effects of the increase in beer price.

- Finding EV is similar to finding income-effect
- If substitution-effect  $\neq 0$ , then  $EV > \text{tax revenue}$

# Excess Burden at individual level



# Excess Burden at individual level



# Efficiency effects of taxation

This helps to understand Excess Burden:

Beer tax increases price of beer, which has two effects

1. Consumer can buy less goods (**income-effect**)
2. Relative price of beer increases → consumer shifts consumption away from beer towards other, non-taxed goods (or leisure) (**substitution-effect**)  
→ This shift makes that tax revenue < utility loss

In other words:

Through substitution effect, taxes **distort behaviour**

These distortions lead to Excess burden.

# Efficiency effects of taxation

Do all taxes have an excess burden?

Definition:

A tax without excess burden is a **Lump-sum tax**

Equivalent definitions:

- Lump-sum tax does not distort behaviour
- With a Lump-sum tax, substitution effect = 0

# Efficiency effects of taxation

Is an income tax a lump-sum tax?

Answer: **no!**

→ Income can be affected: choice between income and leisure

If  $Y$  = income;  $T$  = max hours;  $w$  = wage, then

Budget constraint without income tax

$$Y = [T - L]w$$

Budget constraint with (proportional) income tax  $t$

$$Y = [T - L](1 - t)w$$

→ The higher  $t$ , the more attractive leisure relative to work

# Efficiency effects of taxation

Example of a lump-sum tax:

**A ‘Head tax’**

(If you have a head, you pay  $X$  euro in taxes;

→ Impossible to reduce taxes through change in behaviour )

Main reason for not implementing head taxes?

# Efficiency effects of taxation

Exercise:

$$U = \sqrt{C} + 4\sqrt{B};$$

$$\text{Budget} = 90; P_C = 1; P_B = 2; t_B = 2$$

Supply of B perfectly elastic

Determine excess burden of tax  $t_B$  at ind. level.

# Concluding remarks

Important to remember

- Statutory incidence (who pays) has nothing to do with who bears tax burden → tax shifting
- Economic incidence depends on elasticities
- The more elastic, the lower the share of the tax burden
- When taxes distort behaviour (and almost all taxes do), then tax revenue < loss in efficiency: → Excess Burden
- If no distortion in behaviour: ‘lump-sum tax’

Next week: Optimal taxation

- Minimising (total) excess burden
- Taking distributional concerns into account